

USED MEDIAS LLC

EMBEDDED SYSTEMS DIVISION

Engineering Documentation

cyd-os

DOCUMENT	REVISION	STATUS	CLASSIFICATION
cyd-os	—	—	Internal

Purpose of this set

These reports record the design of cyd-os in enough detail that the project can be picked up cold — by someone else, or by the author after six months away — without re-deriving decisions from the source.

Each report states **what was decided, what it was measured against, and what remains unverified**. Claims that were tested on hardware are marked as such. Claims taken from documentation or reasoning are marked separately, because on a from-scratch kernel the difference between "this is true" and "this should be true" is the difference between a working boot and a silent reboot.

Index

ID	Title	Covers
UM-CYDOS-001	System Architecture and Scope	Layer model, what is built vs borrowed, the isolation problem, why a bytecode VM
UM-CYDOS-002	Boot Chain and Image Format	ROM → 2nd stage → kernel, image header, measured boot trace
UM-CYDOS-003	Xtensa ABI Selection	Windowed vs call0, codegen evidence, consequences for the scheduler
UM-CYDOS-004	Memory Map and Allocation Policy	ESP32 regions, our layout, the bootloader overlap risk
UM-CYDOS-005	Build and Flash Pipeline	Toolchain, flags and why each one, reproducing a build, why not PlatformIO
UM-CYDOS-006	Milestone 0 Verification Report	Test method, captured output, pass/fail per assertion
UM-CYDOS-007	Development Roadmap M1–M5	Each milestone, its risks, and its exit criteria
UM-CYDOS-008	Milestone 1 Verification Report	Vectors, timer source and level choice, interrupt entry/exit, measured results
UM-CYDOS-009	Milestone 2 Verification Report	Task model, context frame, scheduler, the zero-overhead <code>LOOP</code> defect, watchdog correction
UM-CYDOS-010	Milestone 3 Verification Report	Heap allocator, arena model and bounds checking, and the measured DRAM budget
UM-CYDOS-011	Flash Cache and Read-Only Data Placement	Mapping <code>.rodata</code> into flash, the 0x20 page congruence, and the cache-off hazard
UM-CYDOS-012	Milestone 4 Verification Report	Register-based bytecode VM, the ISA, the assembler, containment of malformed programs, and the syscalls added since

ID	Title	Covers
UM-CYDOS-013	Milestone 5 Verification Report	Application table and lifecycle, three-level scheduling, the shell, the escape attempt, and messaging without shared memory
UM-CYDOS-014	Locking Primitives	Critical sections vs blocking mutex, task blocking and idle, and a measured starvation defect
UM-CYDOS-015	Display Driver	ILI9341, span rendering with no framebuffer, and the path from bit-banged SPI to SPI2 with DMA
UM-CYDOS-016	Display Syscalls, and a Total System Freeze	Per-application viewports, pointer discipline, measured containment, and a yield that stopped the clock
UM-CYDOS-017	Touchscreen, and a Verification Method That Failed Three Times	XPT2046 over PENIRQ, the GPIO two-bank bug, calibration by controlled input, a capture that erased its own evidence, and input confinement

Reading order

New to the project: **001** → **002** → **004** → **003**. That gives the shape of the system, how it starts, where things live, and why the calling convention is unusual.

Picking up implementation work: **006** (what is known-good) then **007** (what is next and what it will break).

Reproducing a build: **005** alone is sufficient.

Status summary as of this revision

Item	State
Milestone 0 — kernel boots, self-checks pass	Complete, verified on hardware
Milestone 1 — timer interrupt, tick counter	Complete, verified on hardware
Milestone 2 — native task switching	Complete, verified on hardware — 3,400+ switches, zero corruption
Milestone 3 — heap and arena model	Complete, verified on hardware — all three exit criteria
Milestone 4 — bytecode interpreter	Complete, verified on hardware — program runs, six fault classes contained
Isolation	Live — every VM memory access bounds-checked, UM-CYDOS-012 §6.2
Full stack	Running — bytecode hosted in a preemptible native task, 3.3M instructions across 450 preemptions with zero accounting drift, UM-CYDOS-012 §6.6
Milestone 5 — multiple applications	Complete, verified on hardware — all three exit criteria
Roadmap M0–M5	Complete. Six native tasks, three scheduling levels, a shell, and an application deliberately written to escape its arena that could not
Locking	Complete — critical sections and a blocking mutex; heap and console both arbitrated, UM-CYDOS-014
Task blocking	Live — <code>TASK_BLOCKED</code> plus an idle task using <code>WAITI</code> , UM-CYDOS-014 §3
Display	Working on hardware — ILI9341, no framebuffer, 43 ms full-screen fill via SPI2 + DMA, UM-CYDOS-015 §5.5
Application graphics	Live — <code>FILL / TEXT / DIMS</code> confined to per-application viewports; 0 escapes across 136 audited fills, UM-CYDOS-016 §5
Touch	Working on hardware — XPT2046 via PENIRQ, both axes calibrated by measurement, UM-CYDOS-017
Application input	Live — <code>SYS TOUCH</code> confined to the asking application's viewport; 81 delivered, 109,211 withheld, 0 confinement failures, UM-CYDOS-017 §8
Application messaging	Live — copied through a kernel mailbox, never shared memory; 278 sent / 277 delivered / 0 bad buffers, UM-CYDOS-013 §8
Application bitmaps	Live — <code>SYS BLIT</code> , arena-bounded source and viewport-clipped destination, UM-CYDOS-012 §10
DRAM budget	Measured — 158,000 B allocatable today (167,680 before <code>TASK_MAX</code> rose to 8); a full framebuffer is unnecessary, UM-CYDOS-010 §7.2

Item	State
Flash cache	Enabled — <code>.rodata</code> mapped from flash, UM-CYDOS-011
Version control	Initialised 2026-08-14
JTAG debug probe	Ordered, not in hand
Bootloader IRAM overlap	Closed — investigated and disproved, UM-CYDOS-004 §5
Panic handler	Closed — exercised deliberately with an <code>l11</code> instruction; prints EXCCAUSE/EPC and halts
Watchdogs	Closed — measured armed, now disabled and read back, UM-CYDOS-009 §8

Standing rule for interrupt handlers — clear `PS.EXCM` before calling C. Hardware sets `EXCM` on interrupt entry, and while it is set the Xtensa zero-overhead loop-back is disabled: a hardware loop body executes once and falls through to whatever sits at `LEND`. GCC emits these loops for ordinary counted C loops, so this silently degrades **every C function reachable from a handler** — drivers and the VM interpreter included, not just the scheduler where it was found. `_handler_level3` now clears it; any future handler must do the same. UM-CYDOS-009 §6.

Standing rule for the scheduler — a yield must never defer the clock it depends on. `task_yield()` originally wrote `CCOMPARE1 = ccount + 64` unconditionally, so any loop yielding faster than 64 cycles pushed the deadline ahead of `CCOUNT` forever, the timer interrupt stopped firing, and the whole kernel froze — the tick is what drives every context switch. Any routine adjusting a scheduler deadline must only ever move it **earlier**. UM-CYDOS-016 §3.

Standing rule for verification — when an instrument reports its own reading invalid, that outranks every value printed beside it. Three times across two sessions a frozen marker, a sample counter stuck at an identical value, and a boot banner in a serial capture each said "this measurement is not what you think", and the plausible-looking numbers next to them were believed anyway. Two conclusions were published and later retracted as a result. Latch the quantity so timing cannot lie about it, then feed the system a controlled input rather than interpreting an uncontrolled one. UM-CYDOS-017 §6.